

ZS-A23

Dimension-Weighted Mediator Semigroups and Observer-Algebra Weights in a Finite-Register de Sitter Model

An exact three-sector Markov-operator correspondence: a Mediator-Graph algebra-generation theorem, a dimension-weighted mediator semigroup unifying the trace weights, the inter-sector rates and the relaxation spectrum, a modular detailed-balance theorem, and the late-time vacuum-scale reduction in Z-Spin Cosmology

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Theme / Code: Astrophysics — **ZS-A23 v3.3**

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Repository: github.com/KennyKang-git/zspin

Verification: 26/26 internal consistency checks PASS | Zero Free Parameters. Built on exact symbolic computation: the Mediator-Graph algebra generation to M_{11} ; the dimension-weighted semigroup (stationary (3,2,6)/11, the general characteristic polynomial $\lambda(\lambda+b\kappa^2)(\lambda+D\kappa^2)$, the spectrum $\{0, -2A/Q, -A\}$, both rates as matrix elements); the sector-covariant uniqueness lemma (commutant 1-dimensional); the exact H-theorem; modular detailed balance $\ln(q_i \square / q \square_i) = -\Delta K$; the observer size-bias $\omega = \text{sizebias}(\pi)$; and the relaxation-ratio and reverse-rate predictions. The audit is attached (zs_a23_v3_3_audit.py); the cosmological identifications (offset, embedding, period-2) remain theorem-level arguments, and the NOTE says so. Independent rerun recommended.

Sole geometric inputs: $A = 35/437$, $Q = 11$, $(Z, X, Y) = (2, 3, 6)$, **LOCKED**, with $\kappa^2 = A/Q = 35/4807$ (PROVEN, ZS-A19). Units: reduced Planck mass $\tilde{M}_P^2 = (8\pi G)^{-1}$. $\tilde{H}_0/\tilde{M}_P \approx 6.0 \times 10^{-61}$ is a disclosed external normalisation, not derived.

This is v3.3, the final patched version (twentieth named version). It applies the five external-review patches to v3.2: (1) Main Theorem 2 is split into a **mathematical** semigroup theorem (PROVEN under edge-isotropy) and a **Z-Spin realization** (DERIVED-CONDITIONAL on the sector-isotropic mediation principle); (2) a **sector-covariant uniqueness lemma** (Schur) shows the dimension-weighted generator is the unique sector-covariant edge-isotropic generator; (3) the rate **units** are fixed by a reference scale Γ_0 (the values $2A/Q$, $6A/Q$ are Γ_0 -normalized dimensionless rates); (4) the **H-theorem** is given an exact one-line proof (the Monte-Carlo is demoted to a regression test); (5) the embedding **domain** is corrected to the face algebra $A_{\text{face}} = \mathbb{C}^{121}$. Zero free parameters; $(A, Q, \dim Z) = (35/437, 11, 2)$ **LOCKED**.

§0. Abstract

ZS-A23 establishes an exact correspondence between the Z-mediated finite-register operator algebra and a reversible three-sector Markov semigroup, and applies it to reduce ZS-A22's late-time vacuum-scale debts.

Main Theorem 1 (Mediator-Graph Algebra Generation, PROVEN). $M_3 \oplus M_2 \oplus M_6$ with nonzero $X-Z$ and $Z-Y$ intertwiners (no $X-Y$ generates $M_{11}(\mathbb{C})$; unique trace (3, 2, 6)/11.

Main Theorem 2 (Dimension-Weighted Mediator Semigroup). (2a, PROVEN) Under edge-isotropic microtransition amplitudes, the generator $q_i \rightarrow \square = \Gamma_0 \kappa^2 d\square$ on the path $X-Z-Y$ has stationary $\pi =$

(3,2,6)/11, characteristic polynomial $\lambda(\lambda+b\kappa^2)(\lambda+D\kappa^2)$ (general in (a,b,c)), spectrum $\{0, -2A/Q, -A\}$ = the ZS-Q7 cubic, and matrix elements $2A/Q, 6A/Q$ = the ZS-M43 rates; a GKLS construction gives it with no energy-ladder assumption, and a uniqueness lemma fixes it as the unique sector-covariant edge-isotropic generator. **(2b, DERIVED-CONDITIONAL)** Its Z-Spin realization is conditional on the sector-isotropic mediation principle.

Main Theorem 3 (Modular Detailed Balance & Observer Size-Bias, PROVEN). $\ln(q_i \rightarrow \square / q \square \rightarrow i) = -\Delta K_i \rightarrow \square$; $\omega = \text{sizebias}(\pi)$, $h_i = 11d_i/49$. Corollary: $D(r|\pi)$ Lyapunov (exact proof); $\gamma_{\text{fast}}/\gamma_{\text{slow}} = Q/2 = 11/2$; reverse rates $3A/Q, 2A/Q$.

Main Theorem 4 (Modular-Centralizer Observer Theorem, PROVEN). ω_{reg} 's modular centralizer is exactly $N = M_3 \oplus M_2 \oplus M_6$, $\sigma_t^\omega(C_i \square) = (d_i/d\square)^{\{it\}} C_i \square$, $e^\omega = 121/49$, index = 3 (tight).

Main Theorem 5 (boundary non-predictivity, PROVEN-with-hypotheses). One free vacuum-proportional counterterm makes the Brown–York coefficient scheme-dependent.

The cosmological reductions. Offset = $A_{\text{dS}}/4G$ (CLPW matching), residual = B3; b-energy DERIVED-CONDITIONAL (carrier = face algebra, combinatorial-holography prediction); embedding Φ_{face} OPEN (needs ZS-F2); period-2 OPEN, sharpened by the two-edge-mediation route. **Net.** The trace weights, the inter-sector rates, the relaxation spectrum, and the modular differences are unified in one dimension-weighted generator, with new exact predictions; the cosmological-constant problem is localized to the embedding, the B3 scale, and the absolute $O(1)$ factor. Zero free parameters; **(A, Q, dim Z) = (35/437, 11, 2) LOCKED.**

Epistemic Status Legend

Table 1. Epistemic status legend.

STATUS	DEFINITION
PROVEN	Mathematical theorem; standard mathematics alone, machine-verifiable.
PROVEN-with-hypotheses	Theorem under explicitly stated physical hypotheses (named in situ).
IMPORTED-PROVEN	Result proved externally, used without re-proof; full citation given.
DERIVED	Z-Spin action plus standard physics; zero free parameters.
DERIVED-dynamical	DERIVED from an explicit open-system / dynamical mechanism (leading order).
DERIVED-CONDITIONAL	DERIVED conditional on listed hypotheses not themselves closed.
PREDICTION	A falsifiable consequence stated as such.
HYPOTHESIS-strong / weak	Motivated conjecture; documented closure route / failing chain.
OBSERVATION / CONSISTENCY	A numerical agreement reported as such, not a derivation.
NON-CLAIM	Explicit declaration of what is NOT asserted.
OPEN	Recognized gap; an executable closure route is named where one exists.
CLOSED-NEGATIVE	A registered gate / sub-route resolved in the negative (a theorem).
WITHDRAWN	A prior-version claim retracted as unsupported, with reason.
LOCKED	Core constant fixed upstream; immutable downstream.

§1. Introduction

The ZS-A23 program reduces ZS-A22's P0 (action-level positive vacuum source) and B3 (absolute IR scale) debts at their true level. Through nineteen versions it accumulated a set of operator-algebra and

dynamical results that were, until now, separate: the sector trace weights (3, 2, 6)/11 (A23.5a), the inter-sector relaxation rates $2A/Q$, $6A/Q$ (ZS-M43), the ZS-Q7 relaxation cubic, the modular difference $-\ln 2$ (ZS-F19), and the observer weight (9, 4, 36)/49 (A23.MC).

v3.2 unified all of these; v3.3 seals the result for external review. The organizing insight — that every allowed microstate transition carries the same Z-Spin mediation strength $\kappa^2 = A/Q$, so the coarse-grained sector rate is $q_i \rightarrow \square = \Gamma_0 \kappa^2 d\square$ (one quantum of mediation per available final microstate) — fixes a single dimension-weighted mediator generator whose stationary distribution is the trace weights, whose spectrum is the ZS-Q7 cubic, whose matrix elements are the ZS-M43 rates, whose log-ratios are the ZS-F19 modular differences, and whose operator-space size-bias is the A23.MC weight. The energy-ladder reading of v3.0/v3.1 is removed by a GKLS construction; a uniqueness lemma fixes the generator; and the result is exact and general (the characteristic polynomial factors as $\lambda(\lambda+b\kappa^2)(\lambda+D\kappa^2)$ for arbitrary sector dimensions), so it is structural, not a numerical coincidence — yielding new exact predictions (the relaxation ratio $Q/2$, the reverse rates).

§§3–6 give the five main theorems; §7 upgrades the A23.8 retraction to the dimension-weighted-Laplacian theorem and gives a quantum-simulator protocol; §8 carries the cosmological reductions; §§9–12 the cross-version audit, ledger, conclusion, and residuals; Appendices A (Mediator-Graph proof), B (the generator's exact spectrum). The posture is the corpus's honest conditional reduction.

§2. Locked Inputs and Conventions

(A, Q, (Z,X,Y)) = (35/437, 11, (2,3,6)) LOCKED. $\kappa^2 = A/Q = 35/4807$, $1/\kappa^2 = Q/A = 4807/35$ (PROVEN, ZS-A19). Units: reduced Planck mass $\bar{M}_P^2 = (8\pi G)^{-1}$; $\bar{H}_0/\bar{M}_P = 5.9 \times 10^{-61}$ (NON-CLAIM, = B3). Dynamical algebra $A_{\text{phys}} = M_{11}(\mathbb{C})$ (Hilbert dim 11, operator dim 121); observer algebra $N = M_3 \oplus M_2 \oplus M_6$ (operator dim 49). **121 = dim M_{11} is an operator count; rank $P \leq 11$ in M_{11} .** Sector dimensions $(d_X, d_Z, d_Y) = (a, b, c) = (3, 2, 6)$, $D = a + b + c = Q = 11$. Mediator-Graph = path $X-Z-Y$ (A23.7); $L_{XY} \equiv 0$. Mediation strength κ^2 is the single dynamical coupling.

Rate units (patch). $\kappa^2 = A/Q$ is dimensionless. Rates are stated in the **dimensionless-time convention** $\tau = \Gamma_0 t$, in which $q_i \rightarrow \square(\tau) = \kappa^2 d\square$; the physical rate is $q_i \rightarrow \square(t) = \Gamma_0 \kappa^2 d\square$, with Γ_0 the (single) microscopic attempt scale. The values $2A/Q$ and $6A/Q$ are accordingly Γ_0 -normalized dimensionless rates, not dimensionful frequencies.

§3. Main Theorem 1 — Mediator-Graph Algebra Generation

Theorem A23.5a / A23.7 (PROVEN — Appendix A). Blocks $M_{\{d_i\}}(\mathbb{C})$ plus one nonzero intertwiner per edge generate, within each connected component α , the full $M_{\{D\alpha\}}(\mathbb{C})$ ($D\alpha = \sum_{i \in \alpha} d_i$); the generated algebra is $\oplus_{\alpha} M_{\{D\alpha\}}(\mathbb{C})$, irreducible ($= M_D$) iff connected, with $\dim(\text{commutant}) = \#\text{components}$. **Z-Spin is the path $X-Z-Y \rightarrow M_{11}(\mathbb{C})$** , unique trace $\tau(P_X, P_Z, P_Y) = (3, 2, 6)/11$. Verified by the matrix-unit argument, a multiplicity-2 control, and a span closure to 121 ($= \dim M_{11}$; the Hilbert dimension is 11). The path is connected and ordered — the connectivity gives the algebra (this theorem), the ordering gives the semigroup (§4).

§4. Main Theorem 2 — The Dimension-Weighted Mediator Semigroup

The principle. Every allowed microstate transition between adjacent sectors carries the same Z-Spin mediation strength $\kappa^2 = \mathbf{A}/\mathbf{Q}$. A microstate in sector i can transition to any of the $d_\square$ microstates of an adjacent sector j , so the coarse-grained sector-population rate is $q_{i \rightarrow \square} = \Gamma_0 \kappa^2 d_\square$ (one mediation quantum per available final microstate), with no X–Y transition ($L_{XY} \equiv 0$). On the path X–Z–Y this fixes the generator $\mathbf{Q}$ (rows = rate $i \rightarrow j$; $\Gamma_0 = 1$ in dimensionless time):

$$\mathbf{Q} = \kappa^2 \cdot \begin{bmatrix} [-d_Z, d_Z, 0], [d_X, -(d_X+d_Y), d_Y], [0, d_Z, -d_Z] \end{bmatrix} = \kappa^2 \cdot \begin{bmatrix} [-2, 2, 0], [3, -9, 6], [0, 2, -2] \end{bmatrix}.$$

§4.1 A23.14a — Mathematical Semigroup Theorem [PROVEN]

Theorem A23.14a (PROVEN). For the path X–Z–Y with edge-isotropic rates $q_{i \rightarrow \square} = \Gamma_0 \kappa^2 d_\square$ and sector dimensions (a, b, c):

(i) **Stationary distribution.** $\pi \mathbf{Q} = 0$ has the unique solution $\pi_i = \mathbf{d}_i / \mathbf{D} = (3, 2, 6) / 11$ — the A23.5a trace weights, now the dynamical stationary distribution.

(ii) **Exact spectrum (general).** The characteristic polynomial factors, for arbitrary (a, b, c), as

$$\det(\lambda \mathbf{I} - \mathbf{Q}) = \lambda(\lambda + b\kappa^2)(\lambda + \mathbf{D}\kappa^2),$$

so the eigenvalues are $\{0, -b\kappa^2, -\mathbf{D}\kappa^2\}$. With $b = d_Z = 2$, $\mathbf{D} = \mathbf{Q} = 11$, $\kappa^2 = \mathbf{A}/\mathbf{Q}$: $b\kappa^2 = 2\mathbf{A}/\mathbf{Q}$ and $\mathbf{D}\kappa^2 = \mathbf{Q} \cdot (\mathbf{A}/\mathbf{Q}) = \mathbf{A}$, giving $\{0, -2\mathbf{A}/\mathbf{Q}, -\mathbf{A}\}$ — exactly the ZS-Q7 cubic $\lambda(\lambda + 2\mathbf{A}/\mathbf{Q})(\lambda + \mathbf{A}) = 0$ (verified by exact symbolic computation, both for (3, 2, 6) and for symbolic (a, b, c)).

(iii) **Both corpus rates as matrix elements.** $q_{\square \{X \rightarrow Z\}} = d_Z \kappa^2 = 2\mathbf{A}/\mathbf{Q}$ and $q_{\square \{Z \rightarrow Y\}} = d_Y \kappa^2 = 6\mathbf{A}/\mathbf{Q}$ — the two ZS-M43 rates are entries of the generator, not separate inputs. These are mathematical identities, given the generator; the Monte-Carlo H-theorem of v3.2 is replaced by the exact proof of §5.

The GKLS construction (no energy ladder). Define, for each allowed edge $i \sim j$ and microstates $\alpha = 1 \dots d_i$, $\beta = 1 \dots d_\square$, the jump operator $\mathbf{L}_{\{j\beta \leftarrow i\alpha\}} = \sqrt{(\Gamma_0 \kappa^2)} |j, \beta\rangle \langle i, \alpha|$ (and $L_{\{Y \leftarrow X\}} = 0$). The GKLS generator, summed over sector populations, gives $dp_i/dt = \sum_{\{j \sim i\}} \Gamma_0 \kappa^2 (d_i p_\square - d_\square p_i)$, i.e. exactly $q_{i \rightarrow \square} = \Gamma_0 \kappa^2 d_\square$ — with no energy ladder, no equal-spacing assumption, and no bath spectral density; $L_{XY} = 0$ reflects the absent X–Y intertwiner directly.

§4.2 A23.14c — Sector-Covariant Uniqueness Lemma [PROVEN]

Lemma A23.14c (PROVEN). Among GKLS generators that are (i) covariant under $U(d_X) \times U(d_Z) \times U(d_Y)$, (ii) free of any direct X–Y channel, (iii) edge-isotropic (one common mediation scale), and (iv) primitive and trace-preserving, the dimension-weighted generator is unique.

Proof. The jump operators on an edge (i, j) live in $\text{Hom}(H_j, H_i)$, on which $U(d_i) \times U(d_j)$ acts as the irreducible external tensor product fund $\boxtimes$ antifund; its commutant is one-dimensional (verified: the group generates the full matrix algebra $M_{\{d_i d_j\}}$, so the commutant has dimension 1 for both the X–Z and Z–Y edges). By Schur's lemma the (covariant, positive) Kossakowski block is a scalar, so the jump amplitudes are uniform on each edge; edge isotropy sets the common scale to $\Gamma_0 \kappa^2$, giving $q_{i \rightarrow \square} = \Gamma_0 \kappa^2 d_\square$. This seals the “arbitrary generator” objection: under sector covariance and edge isotropy the generator is forced.

§4.3 A23.14b — Z-Spin Realization [DERIVED-CONDITIONAL]

A23.14b (DERIVED-CONDITIONAL). The identification of this generator as the physical Z-Spin sector dynamics is conditional on the **sector-isotropic mediation principle**: that every allowed microtransition carries the common amplitude $\sqrt{\Gamma_0 \kappa^2}$. The GKLS construction (A23.14a) proves this generator is realizable, and the uniqueness lemma (A23.14c) proves it is the only sector-covariant edge-isotropic option; what is not proved here is that the Z-Spin master action selects sector covariance and edge isotropy at the microscopic level. [A23.14a (the mathematics) and A23.14c (uniqueness under symmetry) are PROVEN; A23.14b (the physical Z-Spin realization) is DERIVED-CONDITIONAL on the sector-isotropic mediation principle. The inter-sector rates, stationary trace weights, and full spectrum follow from the single coupling $\kappa^2 = A/Q$ with no energy-ladder hypothesis; the general (a,b,c) factorization shows the slow mode = $d_Z \cdot \kappa^2$ and fast mode = $Q \cdot \kappa^2$ are structural, not numerological.]

§5. Main Theorem 3 — Modular Detailed Balance and Observer Size-Bias

Theorem A23.15 (PROVEN). The dimension-weighted generator of §4 is reversible, and:

(i) **Detailed balance.** $\pi_i q_{i \rightarrow \square} = (d_i/D) \kappa^2 d_{\square} = \kappa^2 d_i d_{\square}/D$, symmetric in i, j ; hence $\pi_i q_{i \rightarrow \square} = \pi_{\square} q_{\square \rightarrow i}$.

(ii) **Modular detailed balance.** With the modular Hamiltonian $K_i = -\ln \pi_i$,

$$q_{i \rightarrow \square} / q_{\square \rightarrow i} = d_{\square} / d_i = \pi_{\square} / \pi_i = e^{\{-(K_{\square} - K_i)\}}, \text{ i.e. } \ln(q_{i \rightarrow \square} / q_{\square \rightarrow i}) = -\Delta K_{i \rightarrow \square}.$$

This single identity links A23.5a's trace weights, A23.MC's modular differences, ZS-F19's $-\ln 2$ ($q_{Z \rightarrow Y} / q_{Y \rightarrow Z} = d_Y / d_Z = 3 \Rightarrow \Delta K_{Z \rightarrow Y} = -\ln 3$; $q_{X \rightarrow Z} / q_{Z \rightarrow X} = 2/3$), ZS-M43's rates, and ZS-Q7's arrow-of-time relaxation into one modular-thermodynamic relation.

(iii) **Observer size-bias.** The observer weight is the size-bias of the stationary distribution: $\omega_i = d_i \pi_i / \Sigma d_{\square} \pi_{\square} = d_i^2 / \Sigma d_{\square}^2 = (9, 4, 36)/49$, with Radon–Nikodym density $h_i = \omega_i / \pi_i = D d_i / \Sigma d_{\square}^2 = 11 d_i / 49$ — exactly the A23.MC density. State sampling gives $\pi \propto d_i$; operator sampling gives $\omega \propto d_i^2$. The two weightings are one Z-mediated equilibrium sampled in state space and in operator space.

Corollary A23.15.1 (exact H-theorem, PROVEN). With detailed balance, write $x = r_i q_{i \rightarrow \square}$, $y = r_{\square} q_{\square \rightarrow i}$. Then

$$d/dt D(r|\pi) = -1/2 \sum_{i,j} (r_i q_{i \rightarrow \square} - r_{\square} q_{\square \rightarrow i}) \cdot \ln[(r_i q_{i \rightarrow \square}) / (r_{\square} q_{\square \rightarrow i})] = -1/2 \sum (x - y) \ln(x/y) \leq 0,$$

since $(x - y) \ln(x/y) \geq 0$ for all $x, y > 0$. Hence $D(r|\pi)$ is a **Lyapunov function** (verified symbolically; the Monte-Carlo over random distributions is retained only as a regression test). The nonzero relaxation rates are $\gamma_{\text{slow}} = 2A/Q$ and $\gamma_{\text{fast}} = A$, giving the new exact prediction $\gamma_{\text{fast}} / \gamma_{\text{slow}} = A / (2A/Q) = Q/2 = 11/2$, with timescales $\tau_{\text{slow}} = Q/(2A)$, $\tau_{\text{fast}} = 1/A$.

The new predictions (PREDICTION). Beyond reproducing $\gamma_{xz} = 2A/Q$ and $\gamma_{zy} = 6A/Q$, the generator fixes the reverse rates $\gamma_{zx} = d_X \kappa^2 = 3A/Q$ and $\gamma_{yz} = d_Z \kappa^2 = 2A/Q$, so the four edge-direction rates are $\gamma_{xz} : \gamma_{zx} : \gamma_{zy} : \gamma_{yz} = 2 : 3 : 6 : 2$, with detailed-balance ratios 2/3 and 3. These two reverse rates and the ratio $Q/2$ are new, falsifiable consequences not present in the prior corpus. [PREDICTION.]

§6. Main Theorem 4 — The Modular-Centralizer Observer Theorem

Theorem A23.MC (PROVEN). $\omega_{\text{reg}}(P_i) = d_i^2/49 = (9, 4, 36)/49$ (= the §5(iii) size-bias), density $h = \oplus_i (11 d_i/49) I_{\{d_i\}}$; since the d_i are distinct, (i) the modular centralizer $M^{\wedge}\{\sigma^{\wedge}\omega\} = \text{commutant}(h) = \mathbf{N}$; (ii)

$\sigma_t^{\wedge\omega}(C_i) = (d_i/d)^{\wedge\{it\}} C_i$; (iii) $e^{-c} = 121/49 = \dim M_{11}/\dim N$, $\Delta K_{\{X \rightarrow Y\}} = -\ln 2$. Pinching index = 3 (tight), distinct from 121/49. Where §5 derives ω dynamically (the size-bias of the semigroup stationary state), A23.MC derives it modular-theoretically (the intrinsic centralizer of the regular state) — the same (9, 4, 36)/49 from two routes. A23.5b is a DERIVED observer-algebra realization, conditional only on the entropy offset (now = $A_{dS}/4G$, §8).

§7. The Dimension-Weighted Laplacian (Upgrading A23.8) and a Quantum-Simulator Protocol

The A23.8 upgrade (CLOSED-NEGATIVE $\rightarrow$ general theorem). A23 once claimed the ordinary unweighted graph Fiedler eigenvalue equalled $2A/Q$, then retracted it (the ordinary path-Laplacian Fiedler value is 1, not $2A/Q$). v3.3 explains why: the ordinary graph Laplacian is the wrong generator. The correct object is the dimension-weighted mediator Laplacian $q_i = \kappa^2 a_i d_i$, reversible w.r.t. $\pi_i = d_i/D$; its symmetrization $S = \Pi^{\wedge\{1/2\}} Q \Pi^{\wedge\{-1/2\}}$ has off-diagonal entries $S_{ij} = \kappa^2 a_{ij} \sqrt{(d_i d_j)}$, and its spectral gap is $2A/Q$ — not the ordinary Fiedler value. [The retraction is upgraded to a theorem: the relaxation gap is the dimension-weighted Laplacian gap (incorporating the multiplicities d_i), explaining the failure of the ordinary unweighted graph gap. PROVEN.]

A quantum-simulator protocol (TESTABLE). The architecture $H = \mathbb{C}^3 \oplus \mathbb{C}^2 \oplus \mathbb{C}^6$ (a 3-level X register, a 2-channel Z mediator, a 6-channel Y reservoir) realizes the generator directly: prepare population in X, activate only the X–Z and Z–Y jump channels with equal microtransition amplitudes, measure $p_X(t)$, $p_Z(t)$, $p_Y(t)$, fit the two decay exponents, and check the equilibrium ratio. Predictions: $p(\infty) = (3, 2, 6)/11$, $\lambda_{\text{slow}} = -2A/Q$, $\lambda_{\text{fast}} = -A$, $\lambda_{\text{fast}}/\lambda_{\text{slow}} = Q/2 = 11/2$; observer tomography checks the size-biased $\omega = (9, 4, 36)/49$. [TESTABLE — the core A23 dynamics is verifiable on a small open-system simulator without awaiting the absolute cosmological scale.]

§8. The Cosmological Reductions

(offset) — DERIVED-CONDITIONAL. The CLPW reference matching fixes the crossed-product entropy offset: at the maximum-entropy (tracial) state the Type-II entropy is zero, so the offset equals the de Sitter horizon entropy, $S_0 = A_{dS}/4G = 8\pi^2 \bar{M}_P^2/H^2$ (Gibbons–Hawking) — not free. A23.5b's residual reduces to the corpus-wide **B3 absolute scale** $H/\bar{M}_P$.

(b-energy) — DERIVED-CONDITIONAL. Since $\text{rank } P \leq 11$ in M_{11} , the face counts (6, 32, 83) need a separate 121-carrier = the **face algebra** $A_{\text{face}} = \mathbb{C}^{121}$ (the abelian algebra of functions on the 121 faces; structurally distinct from $L^2(M_{11})$). Within-component slot-universality is DERIVED (polytope face-transitivity), and the cross-component one-quantum-per-face is **combinatorial (not metric) holography** — a PREDICTION, since the faces have unequal areas (cube 1.33 vs dodecahedron 0.88). A BV–BFV/TQFT state-sum route (ZS-F0's cobordism functor) could promote it: Pachner-move invariance with one representation label per face forces equal face weights, so topological state-sum invariance $\Rightarrow$ equal face weight.

(embedding) — OPEN. The trace-preserving *-homomorphism $\Phi_{\text{face}} : A_{\text{face}} \rightarrow M_{dS}$ — with domain the face algebra $A_{\text{face}} = \mathbb{C}^{121}$ (not a Hilbert space) — selecting the rank-(6, 32, 83) face-indicator projectors needs the ZS-F2 §11.4 construction; the matter/empty-cell picture (38 filled + 83 empty cells of the $Q^2 = 121$ register, $\Omega_{\text{cdm}} = 32 = \text{dodecahedron} + \text{icosahedron faces}$) is a coherent interpretation, not a derivation. **The deepest remaining cosmological gap.**

(period-2) — OPEN, sharpened. The absolute $O(1)$ factor $|\Delta W| = 2$: beyond the complex-bion ($S_{\text{bion}} = 2S_{\text{I}}$), RG ($b_0 = 1/2$), and $SU(2)/\mathbb{Z}_2$ routes (all needing the master action), v3.2 adds a Markov large-deviation route built on the §4 generator: the $X \rightarrow Y$ rare-event path must traverse two edges (the graph distance $d(X, Y) = 2$ on the PROVEN path $X \rightarrow Z \rightarrow Y$), so the minimal large-deviation action decomposes $S_{\{X \rightarrow Y\}} = S_{\{X \rightarrow Z\}} + S_{\{Z \rightarrow Y\}}$; if each elementary seam crossing has action $1/\kappa^2$, then $S_{\{X \rightarrow Y\}} = 2/\kappa^2$. **The factor 2 is the mandatory two-edge mediation (a structural graph distance), not a spinor number** — reducing period-2 to the per-edge action normalization. **[OPEN; the double-well control $\Delta W = 4/3$ confirms the value is non-generic.]**

§9. Main Theorem 5 — Finite-Boundary Non-Predictivity

Theorem A23.10 (PROVEN-with-hypotheses). If the admissible finite covariant counterterm class contains one free vacuum-proportional term $\alpha_0 \sqrt{h}$ and no independent normalization fixes α_0 (with boundary diffeomorphism invariance and a finite boundary), then $T_{ij}^{\text{vac}} \mapsto T_{ij}^{\text{vac}} + (\alpha_0/2) \sqrt{h} h_{ij}$. GHY-counterterm-alone route **CLOSED-NEGATIVE**; the Wald first-law fixes α_0 (pure-dS $\rightarrow \Omega_{\Lambda} = 1$). This Brown–York no-go is logically independent of the §4–§6 dynamics; it resolves gate G14c and may be spun off.

§10. Cross-Version Consistency Audit

Table 2. Cross-version dependency audit.

Upstream result	Status	v3.3's effect	Safe?
ZS-A19 $\kappa^2 = A/Q = 35/4807$	PROVEN	§4: the single generator coupling; unchanged	✓
A23.5a weights (3,2,6)/11	PROVEN	§4: now also the dynamical stationary distribution	✓
ZS-M43 rates $2A/Q, 6A/Q$	PROVEN	§4: now matrix elements; reverse rates $3A/Q, 2A/Q$ predicted	✓
ZS-Q7 cubic $\lambda(\lambda+2A/Q)(\lambda+A)$	PROVEN	§4: now the generator's characteristic polynomial	✓
ZS-F19 $\Delta K_{\{X \rightarrow Y\}} = -\ln 2$	DERIVED	§5: from modular detailed balance $\ln(q_i \square/q \square_i) = -\Delta K$	✓
A23.MC $\omega = (9,4,36)/49$	PROVEN	§5(iii): the size-bias of π ; §6: the centralizer — two routes	✓
A23.8 (graph gap)	WITHDRAWN	§7: upgraded to the dimension-weighted Laplacian theorem	✓
ZS-F2 §11.4 $\Omega_{\Lambda} = 83/121$	DERIVED	§8: combinatorial-holography prediction; embedding OPEN	✓
CLPW max-entropy = unit 1	IMPORTED-PROVEN	§8: offset = $A_{\text{dS}}/4G$	✓

Result. No version conflict; v3.3 unifies five previously separate PROVEN/DERIVED results into one generator, seals the realization with a uniqueness lemma, and upgrades the A23.8 retraction, without disturbing anything upstream. **[DERIVED.]**

§11. Consolidated Gate Ledger

Table 3. Consolidated gate ledger (v3.3).

Gate	Calculation	Status (v3.3)
G-algebra (MT1)	$M_3 \oplus M_2 \oplus M_6 + C = M_{11}$; weights (3,2,6)/11.	PROVEN (A23.5a / A23.7).

Gate	Calculation	Status (v3.3)
G-semigroup (MT2)	dimension-weighted generator.	A23.14a PROVEN; A23.14c uniqueness PROVEN; A23.14b realization DERIVED-CONDITIONAL.
G-modular (MT3)	detailed balance & size-bias.	PROVEN (A23.15): exact H-theorem; $Q/2$; reverse rates.
G-centralizer (MT4)	observer = modular centralizer.	PROVEN (A23.MC); index 3.
G12a-phys (A23.5b)	de Sitter observer realization.	DERIVED-CONDITIONAL — offset = $A_{dS}/4G$ (§8); residual = B3.
G12b-exponent	$1/\kappa^2 = Q/A$.	PROVEN.
G12b-O(1)	$S_{inst} = 2/\kappa^2 = \Delta W = 2$.	OPEN; two-edge-mediation makes ‘2’ structural (§8); per-edge action pending.
G14c (MT5)	boundary reproduces Ω_Λ ?	CLOSED-NEGATIVE (GHY-alone); Wald fixes coeff.
G14-energy (b-energy)	$\Omega_i = \text{rank } P_i/121$.	DERIVED-CONDITIONAL — combinatorial-holography PREDICTION (§8); embedding OPEN.
G-laplacian (A23.8)	spectral gap.	PROVEN — dimension-weighted Laplacian gap = $2A/Q$ (§7).

§12. Conclusion

Across twenty versions the ZS-A23 program has matured from a defensive record of “how far the vacuum-scale debt was reduced” into an exact correspondence between a finite-register operator algebra and a reversible three-sector Markov semigroup.

The unification (the centerpiece). One dimension-weighted generator — built from the single coupling $\kappa^2 = A/Q$ by the rule $q_i \rightarrow \square = \Gamma_0 \kappa^2 d\square$ — has the trace weights (3, 2, 6)/11 as its stationary distribution (A23.5a), the cubic $\lambda(\lambda + 2A/Q)(\lambda + A)$ as its characteristic polynomial (ZS-Q7), the rates $2A/Q$, $6A/Q$ as its matrix elements (ZS-M43), the modular difference $-\ln 2$ from its log-ratios (ZS-F19), and the observer weight (9, 4, 36)/49 as its operator-space size-bias (A23.MC). The general (a, b, c) factorization $\lambda(\lambda + b\kappa^2)(\lambda + D\kappa^2)$ shows the slow mode is set by d_Z and the fast mode by Q — structural, not numerological — a GKLS construction removes the energy-ladder assumption, and a Schur uniqueness lemma fixes the generator. New exact predictions follow: the relaxation ratio $Q/2 = 11/2$ and the reverse rates (the four edge rates 2 : 3 : 6 : 2). The A23.8 retraction is upgraded to the dimension-weighted-Laplacian theorem.

The cosmological reductions. The entropy offset is the de Sitter entropy $A_{dS}/4G$ (residual = the universal B3 scale); b-energy is DERIVED-CONDITIONAL with the dark-sector budget identified as a combinatorial-holography prediction (with a BV–BFV closure route); the embedding functor $\Phi_{face} : A_{face} \rightarrow M_{dS}$ (needing ZS-F2) and the absolute O(1) factor remain OPEN, the latter sharpened so its factor 2 is the mandatory two-edge mediation.

Net. The dynamical heart of A23 — trace weights, inter-sector rates, relaxation spectrum, modular differences, observer weight — is now a single dimension-weighted mediator principle with new predictions and an experimental protocol; the cosmological-constant problem is localized to the embedding, the B3 scale, and the absolute O(1) factor. The corpus standard — honest conditional reduction, anti-numerology enforced (the general (a, b, c) factorization, the double-well control), every over-claim retracted — is maintained. Zero free parameters; $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED.

§13. Remaining Programs and Spin-Off Potential

Remaining. (embedding) $\Phi_{\text{face}} : A_{\text{face}} \rightarrow M_{\text{dS}}$ from ZS-F2 (§8). (combinatorial holography) the BV–BFV state-sum giving equal face weights (§8). (B3) the absolute IR scale $H/\bar{M}_P$ (the corpus-wide debt). (period-2) the per-edge large-deviation action normalizing each seam crossing to $1/\kappa^2$ (§8). **Spin-off potential:** (1) the dimension-weighted mediator semigroup (§§4–7) — algebra generation, the exact Markov–operator correspondence, the uniqueness lemma, modular detailed balance, the exact H-theorem, the dimension-weighted Laplacian, and the simulator protocol — is a self-contained operator-algebra / open-quantum-systems paper, the strongest spin-off; (2) the de Sitter observer algebra with the CLPW offset $= A_{\text{dS}}/4G$; (3) the Boundary Non-Predictivity Theorem (§9); (4) the unified spin-graded continuous-clock program (ZS-A24).

Appendix A — General Proof of the Mediator-Graph Theorem

Let A be generated by $\oplus_i M_{\{d_i\}}(\mathbb{C})$ (matrix units $e^{\{(i)\}}_{\{ab\}}$) and, per edge (i, j) , one nonzero $C_i \square$ with a nonzero entry at (a_0, b_0) . Then $e^{\{(i)\}}_{\{a_0\}} \cdot C_i \square \cdot e^{\{(j)\}}_{\{b_0\}}$ is a nonzero multiple of $e_{\{ab\}}$, so $A \supseteq \text{Hom}(H_j, H_i)$ (and adjoints). Iterating along a connected path generates every $e_{\{ab\}}$ within a component, so $A = \oplus_{\alpha} M_{\{D\alpha\}}(\mathbb{C})$, $= M_D$ iff connected, $\dim(\text{commutant}) = \#\text{components}$. The path $X\text{--}Z\text{--}Y$ is connected $\rightarrow M_{11}$, and ordered $\rightarrow$ the §4 semigroup. The 121 is the operator dimension; the Hilbert dimension is 11, rank $P \leq 11$.

Appendix B — The Generator's Exact Spectrum

For the path $X\text{--}Z\text{--}Y$ with $q_i \rightarrow \square = \kappa^2 d \square$, $\mathbf{Q} = \kappa^2 [[-b, b, 0], [a, -(a+c), c], [0, b, -b]]$ (rows = rate $i \rightarrow j$). Direct expansion:

$$\det(\lambda I - \mathbf{Q}) = (\lambda + b\kappa^2) \cdot \{(\lambda + (a+c)\kappa^2)(\lambda + b\kappa^2) - b(a+c)\kappa^4\} = (\lambda + b\kappa^2) \cdot \lambda \cdot (\lambda + (a+b+c)\kappa^2) = \lambda(\lambda + b\kappa^2)(\lambda + D\kappa^2).$$

Eigenvalues $\{0, -b\kappa^2, -D\kappa^2\}$; with $(a, b, c) = (3, 2, 6)$, $\kappa^2 = \mathbf{A}/\mathbf{Q}$: $\{0, -2\mathbf{A}/\mathbf{Q}, -\mathbf{A}\}$. The slow gap is set by the mediator dimension $b = d_Z$; the fast rate by the total $D = \mathbf{Q}$. The five invariants of $N \subset M_{11}$ ($\dim H = 11$, $\dim N = 49$, $\dim M_{11} = 121$, ratio $121/49$, index 3) are unchanged; 121 is the operator dimension, structurally distinct from the face-count total $6 + 32 + 83$.

Acknowledgements and Code Availability

This v3.3 applies five external-review patches to v3.2: the split of Main Theorem 2 into a mathematical semigroup theorem (A23.14a, PROVEN) and a Z-Spin realization (A23.14b, DERIVED-CONDITIONAL); the sector-covariant uniqueness lemma (A23.14c, PROVEN via Schur); the rate-unit convention (Γ_0); the exact one-line H-theorem proof (the Monte-Carlo demoted to a regression test); and the correction of the embedding domain to the face algebra A_{face} . The Dimension-Weighted Mediator Semigroup Theorem unifies the sector trace weights, the inter-sector rates, the relaxation cubic, the modular difference, and the observer weight as exact consequences of one generator built from the single coupling $\kappa^2 = \mathbf{A}/\mathbf{Q}$.

zs_a23_v3_3_audit.py (exact symbolic): algebra generation; the dimension-weighted generator (stationary, the general (a,b,c) factorization, the spectrum, the rates); the **sector-covariant uniqueness lemma** (commutant 1-dimensional); detailed and modular detailed balance; the observer size-bias and RN density; the **exact H-theorem** (with the Monte-Carlo retained as a regression test); the new predictions ($Q/2$, reverse rates 2:3:6:2); the ordinary-vs-dimension-weighted Laplacian comparison; and the

cosmological identifications. The NOTE states: “26/26 internal consistency checks PASS; the cosmological identifications (offset, embedding, combinatorial holography, period-2) remain theorem-level arguments, not code-verified results.” Independent rerun recommended.

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Version History

Table 4. Version history.

Version	Date	Change
v1.0–v2.9	2026	v1.0 over-claimed $\rightarrow$ algebraic weights, Mediator-Graph, observer theorem, Modular-Centralizer unification, programs executed, corrections. Superseded.
v3.0	2026	Four priority programs: the Davies rate $\gamma_i \rightarrow \square = d\square \cdot (A/Q)$; carrier $\neq L^2(M_{11})$; within-component slot-universality; the $\mathbb{Z}_N$ entropy divergence. Superseded.
v3.1	2026	Five residual programs: offset = $A_{dS}/4G$; equal spacing from the path; combinatorial-holography prediction; embedding/period-2 OPEN. Superseded.
v3.2	2026	Restructured around the Dimension-Weighted Mediator Semigroup Theorem; A23.8 upgraded; new predictions $Q/2$, reverse rates 2:3:6:2; quantum-simulator protocol. Superseded.

Version	Date	Change
v3.3	March 2026	Final patched version (twentieth). (1) MT2 split into A23.14a (PROVEN math) + A23.14b (DERIVED-CONDITIONAL realization); (2) A23.14c sector-covariant uniqueness lemma (Schur, PROVEN); (3) rate units fixed by Γ_0 ; (4) exact one-line H-theorem (MC demoted to regression test); (5) embedding domain corrected to $A_{\text{face}} = \mathbb{C}^{121}$. 26/26 checks PASS. Zero free parameters; (A, Q, dim Z) = (35/437, 11, 2) LOCKED. (Consolidated from internal Z-Spin Collaboration research notes up to v1.6.0.)

— End of ZS-A23 v3.3 —